

AcYut, BITS-Pilani's humanoid robot that won the RoboGames last year, is a brilliant example. The RobotCub project (www.robotcub.org) funded by the European Commission to build iCub, a toddler-like humanoid robot, is also based on an open framework. Vineet Sahu, another of the enthusiastic RoboMSR champions, points out that, "The ASIMO which was a closed source project took almost 15 years to develop to the stage that we know it to be today, while the iCub project developed by the European Union as an Open Source project took just six years to finish. This is a perfect example of the difference between closed and Open Source."

Spykee, the network-controllable mobile robotic toy, is also advertised as an Open Source robot which can be integrated with other projects or modified to a user's needs. E-puck, the education robot originally developed at the École Polytechnique Fédérale de Lausanne, Switzerland, for micro-engineering education is also a classic piece of open engineering. The hardware designs as well as the onboard software of the 7cm tall differential-wheeled mobile robot are Open Source, and it is being built and sold by several companies now.

The Humanoid Robot project of Sweden is another Open Source example which has contributed greatly to the development of robot learning techniques. Virginia Tech's Dynamic Anthropomorphic Robot with Intelligence-Open Platform (DARwIn-OP) is a recent example of an open-platform miniature humanoid robot for research, education and outreach. Users are encouraged to modify and use both the hardware and software aspects of DARwIn-OP. CAD files for all of its parts and instructions manuals for fabrication and assembly are available for free on the university's website, and various software implementations are possible using C++, Python, LabVIEW, MATLAB, etc. The platform features advanced computational power, sophisticated sensors, high payload

Just right for learners

Open hardware also happens to be a superb fit for learning kits. Azad explains, "The main aim of learning is to tinker around and see what a student can do with it. If everything is open, it is up to the interests of the students to go as deep as they want, to change the system or develop modules for the existing system. To make this happen, the students will have to understand the entire system and its functioning. Without open systems, every batch of students would do the same task over and over again. With Open Source kits, the students can add value, which would be shared all over the world, and when the next batch comes in, they can pick it up from there. In this process, if anyone wants to go back and change something, the files and data are always available as it is Open Source."

capacity and dynamic motion ability.

All of Willow Garage's projects including PR2 and Turtlebot, are also typical Open Source examples. The robot operating system (ROS), the Open Source computer vision libraries for real-time perception called OpenCV and other components published by Stanford University's Artificial Intelligence Laboratory and Steve Cousins' Willow Garage have become quite popular in the robotics field.

Gostai, a French company that makes surveillance and telepresence robots under its Jazz line of products, is another Open Source success story. With "Robotics for Everyone" as its tagline, the company has not only demonstrated the benefits of using open components but given back to the community, by completely open sourcing its prized piece of technology—Urbi, an Open Source software platform to control robots.

Rohit de Sa points out two other interesting examples: "In the mechanical branch, MakerBot's Thing-O-Matic and RepRap's Mendel are both completely Open Source 3D printers. You can actually print out mechanical parts using melted plastic. Incidentally, the Thing-O-Matic and the Mendel, both use other OSH for their control electronics. So what we're seeing here is a symbiotic relation between multiple branches of engineering. Electronics feeds mechanical, which feeds something else, and so on and so forth."

Take your pick

One of the biggest examples of open hardware used vastly by the robotics community in India is Arduino.

Arduino is an Open Source electronics prototyping platform based on flexible, easy-to-use hardware and software. The Arduino board can receive inputs from a variety of sensors, and in turn control external factors like lights, motors and actuators.

Arduino's reference designs are available on its website as CAD files, and can be modified to suit the user's needs. The board itself can either be bought assembled or built by the user based on the designs. The microcontroller on the board is programmed using the Arduino programming language and the Arduino development environment, both of which can be downloaded for free from the Arduino website www.arduino.cc

"The Arduino platform is creating a revolution in the field of robotics as it is simple to use. The board is also flexible, in the sense that there can be different voltages taken from the same board, there can be digital signals and analogue signals given as inputs, the size of the microcontroller can vary from an ATmega8 to an ATmega2560, and so on. Plus, there are various interfaces for different hardware components starting from the very basic to the more complex. There are also various types of boards available for different utilities. There are many projects that are being carried out with the help of this board for hobbyists as well as for researchers," says Pitchai.

Arduino is quickly gaining traction in India and growing to become the favourite platform for low-end robotics and learning kits. Azad informs that his start-up is also using Arduino widely in its projects: "We are selling Open